TOPIC 05

Reading the Regression Table

The three tables every statistics package prints: model summary, ANOVA and coefficients, and what each number tells you

WHAT YOU WILL LEARN

- Name the three tables in regression output
- Read R^2, adjusted R^2, standard error and observations from the model summary
- Explain the ANOVA table and the F-statistic
- Read the coefficient table: b, standard error, t, p-value, confidence interval
- State and test H_0: $\beta = 0$
- Write the regression equation from the output

NOTE **Where this connects**

Topic 04 defined SST, SSR, SSE and R^2. In practice nobody computes these by hand: software prints them in tables instead of showing a graph. These tables also contain **hypothesis tests and p-values** (Topics 10 and 13 of the inferential course), so everything you learned there is used here.

1. The three tables

Table	What it contains	Question it answers
1. Model summary (Regression Statistics)	Multiple R, R^2, adjusted R^2, standard error, observations	How well does the model fit?
2. ANOVA	df, SS, MS, F, Significance F	Is the model as a whole useful?
3. Coefficients	b, standard error, t, p-value, confidence interval	What are the equation and is each variable significant?

We build all three for the SAT and GPA data (84 students) so that every number can be traced to a formula. Section 5 compares them with the table in the lecture workbook.

2. Table 1: model summary

Item	Value	Meaning
Multiple R	0.6372	Correlation between observed y and fitted $\hat{y}$ (equals $\lvert r \rvert$ in simple regression)
R Square	0.4060	SSR/SST: share of variability explained (Topic 04)
Adjusted R Square	0.3988	R^2 corrected for the number of variables (Topic 07)
Standard Error	0.2106	Typical size of a residual: $\sqrt{(SSE/(n-2))}$, in GPA points
Observations	84	Sample size n

The **standard error of the regression** is in the units of y. Here a typical prediction misses the true GPA by roughly 0.21 points. The lecture says 'the bigger the sample, the lower the error'. That is true of the **standard errors of the coefficients**: larger samples make those estimates

more precise (smaller standard errors in Table 3), but this standard error estimates the noise level σ and does not shrink as n grows. It is not the standard error of a coefficient.

3. Table 2: ANOVA

ANOVA (analysis of variance) is a family of methods for comparing means. Here it simply re-presents SSR and SSE and tests the whole model. The two sums of squares are divided by their **degrees of freedom** to give **mean squares** (MS), and the ratio of those is the F-statistic.

	df	SS	MS	F	Significance F
Regression (SSR)	1	2.4861	2.4861	56.05	7.2e-11
Residual (SSE)	82	3.6373	0.0444		
Total (SST)	83	6.1234			

- **Degrees of freedom (df)** = the number of observations minus the number of parameters estimated. Here two parameters (b_0 and b_1) are estimated, so the residual df is $n - 2$. Regression df is 1 (one predictor); total df is $n - 1$.
- **df check:** regression df + residual df = total df (1 + 82 = 83).
- **MS = SS/df.** MSE is the estimated error variance; its square root is the standard error of the regression above.
- **F = MSR/MSE = 2.4861/0.04436 = 56.05.** A large F means the model explains much more than chance would.
- **Significance F** is the p-value of the F-test of H_0: *the model explains nothing* (all slopes zero). A tiny value means the regression is significant as a whole.
- With one predictor, **$F = t^2$** (the slope's t-statistic squared): $7.4865^2 = 56.05$. The F-test gets more interesting with several predictors (Topic 06).

> **COMMON MISTAKE** **Labels in the lecture slide**
> The ANOVA table in the lecture's regression-table slide labels the first row SSE and the second SSR. They are swapped: the row with df 1 and SS 2.135 is the **regression** (explained, SSR), because $R^2 = SSR/SST = 2.135/5.248 = 0.407$ matches the printed R^2. The Excel workbook labels it correctly. Always read the labels against the formulas.

4. Table 3: coefficients, the heart of the regression

The first row is the **intercept**, the second row the independent variable. Going column by column:

	Coefficient	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	0.2750	0.4087	0.673	0.503	−0.538	1.088
SAT	0.001656	0.000221	7.487	7.2e-11	0.001216	0.002096

1. **Coefficient:** the estimates b_0 and b_1. Together they define the equation (below).
2. **Standard error:** how much each estimate would vary from sample to sample (as with any statistic, Topic 07 of the inferential course). Smaller is better.
3. **t Stat = coefficient/standard error.** How many standard errors the estimate is from zero.

4. **P-value:** the two-sided p-value of the test **H_0: $\beta = 0$** (the coefficient is zero, so the variable has no linear effect on y).
5. **Lower/Upper 95%:** the 95% confidence interval for the coefficient, b ± t* × SE (Topic 09 of the inferential course).

$$\widehat{GPA} = 0.2750 + 0.001656 \times SAT$$

Reading the tests

- **SAT:** t = 7.49, p = 7.2e-11. Far below any usual α, so we **reject H_0: $\beta_1 = 0$**. SAT has a significant linear relationship with GPA. The interval [0.00122, 0.00210] excludes 0, which agrees.
- **Intercept:** p = 0.50 is large, so we cannot say the intercept differs from 0. If the intercept were 0, the line would cross the y-axis at the origin. That matters little here because x = 0 is far outside the data, and we keep the intercept in the model anyway.
- The p-value column is where you hope for tiny numbers (software shows 0.000, which should be reported as p < 0.001). The significance level α is still the researcher's choice.

5. Comparing with the lecture workbook

The lecture workbook (`Regression-table.xlsx`) gives a table for the same kind of data. Its numbers are close to ours but not identical. The table is labelled with 100 observations, while the SAT and GPA data sheet in the lecture has 84 rows. The table's own numbers point to a third size (see the box below), so it was probably produced from a slightly different version of the data.

Item	Lecture workbook	Our fit (84 rows)
R Square	0.407	0.406
Adjusted R Square	0.399	0.399
Standard Error	0.196	0.211
Observations	100	84
F	55.55	56.05
Intercept b_0 (SE, p)	0.4803 (0.3845, 0.215)	0.2750 (0.4087, 0.503)
Slope b_1 (SE, t)	0.0015492 (0.00021, 7.45)	0.0016557 (0.00022, 7.49)

The conclusions are identical: a significant positive slope, an insignificant intercept and R^2 near 0.41. Differences of this size are the kind of variation you expect between two samples (sampling variation, as in the inferential course).

> **NOTE** **Details worth knowing in the workbook table**
>
> - The labels 'Observations 100' and df = 1, 99, 100 do not fit the table's own numbers. Residual df = SSE/MSE = 3.1129/0.03843 = 81, so total df is 82 and n is about 83. The adjusted R^2 of 0.39948 is also reproduced with n = 83: $1 - (1 - 0.40681) \times 82/81 = 0.39948$.
> - The standard error printed on the 'minor formatting' sheet is 0.178, whereas MSE = 0.0384 gives $\sqrt{0.0384} = 0.196$, the value on the slide.
> - The lecture rounds b_1 to 0.0015 in its equation. Use the unrounded 0.0015492 for predictions (Topic 03).

These are small blemishes in the teaching material, not in the method. The way to catch them is the habit of checking: SS + SS = SS, df + df = df, R^2 = SSR/SST, F = MSR/MSE, t = b/SE.

6. A checklist for reading any regression output

1. Look at **R^2 and adjusted R^2**: how much is explained?
2. Look at **Significance F**: is the model useful at all?
3. Read the **coefficients** and write the equation.
4. Check each **p-value** (or interval): which variables are significant?
5. Check the **units** and sign of each coefficient against common sense.
6. Remember: significance is not causation (Topic 01).

7. In Python

```
import numpy as np
import statsmodels.api as sm

data = np.loadtxt("satgpa.csv", delimiter=",", skiprows=1)
X = sm.add_constant(data[:, 0])          # adds the intercept column
model = sm.OLS(data[:, 1], X).fit()

print(model.summary())                    # all three tables
print(model.params, model.bse)            # b and standard errors
print(model.tvalues, model.pvalues)
print(model.conf_int(alpha=0.05))         # 95% intervals
print(model.rsquared, model.rsquared_adj, model.fvalue)
```

SUMMARY

- Regression output has three tables: model summary, ANOVA and coefficients.
- Model summary: R^2, adjusted R^2, standard error of the regression ($\sqrt{MSE}$) and n.
- ANOVA: df, SS, MS, F = MSR/MSE and Significance F, the p-value of the whole-model test. With one predictor $F = t^2$.
- Coefficients: b, SE, t = b/SE, p-value for H_0: $\beta = 0$, and a 95% interval.
- Write the equation from the coefficients; keep enough decimals.
- Always verify that the table is internally consistent.

Practice questions

Try each question before reading the answer underneath it.

Q1. In the coefficient table, SE of the slope is 0.0004 and the slope is 0.0012. Find t.

Answer. t = 0.0012/0.0004 = 3.0.

Q2. A coefficient has p = 0.000 in the output. What does it say about H_0: $\beta = 0$, and how do you report it?

Answer. Reject H_0: the variable is significant. Report $p < 0.001$.

Q3. The ANOVA table has SSR = 30, SSE = 10, n = 12, one predictor. Find F.

Answer. MSR = 30; MSE = 10/10 = 1; F = 30.

Q4. A 95% interval for a slope is [−0.1, 0.4]. Is the variable significant at $\alpha = 0.05$?

Answer. No. The interval contains 0, so H_0: $\beta = 0$ cannot be rejected.

Q5. Write the equation if the coefficients are intercept 1.2 and 'hours' 0.5.

Answer. $\hat{y} = 1.2 + 0.5 \times$ hours.

Q6. Why is an insignificant intercept usually not a reason to drop it from the model?

Answer. It only says the line may pass through the origin. The intercept also positions the line, and $x = 0$ is often outside the data range.